



How do congressional members appear on the web? Tracking the web visibility of South Korean politicians

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ABSTRACT

The web visibility of politicians has substantial implications for the internet and politics. In this regard, the present paper examines the web visibility of South Korea's 18th National Assembly members. This study identifies those members with the highest web visibility and determines the factors affecting their web visibility by using a set of socio-demographic variables indicating web visibility. The concept of web visibility is operationally defined as the number of web mentions of each congressional member. Web mentions of the name of each congressional member were collected across many platforms, including blogs, images, news, and websites, by using a Naver search tool based on the API, and relationships between the web visibility of the members and their socio-demographic attributes (their gender, age, term, constituency, and party affiliation) were analyzed using Kruskal–Wallis, Median, and Mann–Whitney U tests. The most visible politicians in the Korean websphere had highly qualified political careers and prominence, and the difference in politicians' web visibility depended on their political attributes (their term, constituency, and party), not on their demographic attributes (their gender and age). These findings suggest that politicians who build political power and influence through their political career in the offline world are likely to be more visible in the websphere. Further, an integrated indicator based on politicians' web visibility status can reflect their "ubiquitous presence" in Korea's digitalized society.

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1. Introduction

The internet has been used as a key source of information on congressional and presidential campaigns. In 2008, approximately 24% of Americans indicated that they obtained valuable campaign information from the internet; the figure was only 13% in 2004 (Pew Research Center, 2008). This indicates that politicians have been using the internet for communicating with the general public on their political activities (Lipinski & Naddor, 2004). The internet has become firmly established as a means of expanding social networks as well as a communication space for open and interactive communication. It can facilitate the interactive communication process by (re) calling people's attention to central/peripheral, previously highlighted/isolated political actors and their activities (Anstead, 2008). Internet-mediated technologies are expected to further reconfigure the social and political presence and relations (Park & Kluver, 2009).

The relationship between politics and the internet has become more intimate as a result of the popularity of the internet and increasing numbers of innovative web applications. Accordingly, there has been growing research interest in the political use of the

internet; the institutional analysis of the relationship between the internet and politics (Anstead & Chadwick, 2008); the ideological web profiling of political bloggers (Kirchhoff, Nicolai, Bruns, & Highfield, 2009); the policy use of blogs (McKenna, 2007); and the content analysis of politicians' websites (Gulati, 2004).

An important result of the merger between the internet and politics is online political visibility, a key research topic. The internet has become an integral part of everyday political life, and thus, profiling politicians' presence on the internet has become very important. On the other hand, few studies have quantitatively tracked and evaluated the online visibility of politicians across various web platforms. Therefore, the present study provides a statistical analysis of such phenomena by using several web-based metrics. More specifically, to better understand Korean politics, this study examines the web visibility of Korea's 18th National Assembly members by automatically tracking their web visibility.

There were 37.48 million internet users in South Korea (hereafter "Korea") in 2009, accounting for 77.3% of the total population; Korea had the highest internet penetration rate in Asia in 2009 (Internet World Stats, 2009). This indicates that Korea has an extensive range of internet users, which makes Korea an ideal case study. Hara and Jo (2007) examined the U.S. and Korea and suggested that the internet provides political resources and opportunities necessary for ordinary Korean citizens to participate in the political process. Further, the internet has dramatically changed the way political messages are

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conveyed in Korea. This has been mainly because the broadband penetration rate in Korea is approximately four times that in the U.S., which has made the internet a viable alternative medium for political information and communication (Moon & Yang, 2003). Further, Korean politicians have been using the internet not only for news but also for websites, blogs, photos, and videos, among others. This increasing range of internet content indicates that the pattern of internet use among politicians is becoming more volatile.

2. Relevant literature

2.1. Studies of the political role of the internet

In modern democracies, the media have played an important role in connecting political actors with citizens. A number of studies have examined the political influence of the media by using priming theory, framing theory, and agenda-setting theory, among others (Bennett & Iyengar, 2008; Chong & Druckman, 2007; Dumitrescu & Mughan, 2010; Iyengar & Kinder, 1987; McCombs, 2004; Scheufele & Tewksbury, 2006). Arguments based on the media effect have typically focused on media messages and audiences' perceptions, attitudes, and behavior changes.

While these arguments have a micro perspective regarding the relationship between the media and politics, "the mediatization of politics" provides a comprehensive perspective, including media systems, political systems, social cultures, and actors in the process of the media influence on politics (Mazzoleni, 2008; Mazzoleni & Schulz, 1999; Strömbäck, 2008; Strömbäck & Dimitrova, 2011). The mediatization of politics can be defined as a dynamic process in which the media come to assume a central position in the political arena and shape politics and society (Mazzoleni, 2008; Strömbäck, 2008; Strömbäck & Dimitrova, 2011).

Although mediatization theory provides an inclusive and broad understanding of the political influence of the media, it generally focuses on the traditional mass media, such as television, radio, newspapers, and journals. Recently, Tresch (2009) found that the news coverage of Swiss politicians tends to focus on the most active and authoritative politicians. That is, political news reflects political reality and depends on the newsworthiness of the elite status of politicians. Sellers and Schaffner (2007) suggested that the traditional media reproduce the established political structure in the real world.

Schulz (2004) indicated that the internet appears to follow the logic of the traditional media. However, the implications of the internet in terms of the mediatization of politics remain unclear because the internet reflects a wide range of media formats and content produced by many producers, including politicians, citizens, and professional journalists (Strömbäck, 2008). In addition, the internet is constantly developing and expanding, and thus, it is difficult to clearly define its position in society. Strömbäck (2008) indicated that the internet is still in its early stages, and thus, researchers should continue their efforts to examine and monitor the effects of the internet on sociopolitical environments.

Although there is no theory underpinning the political influence of the internet, there have been two competing theoretical scenarios with respect to the overall impact of new internet technologies on politics (Gibson & Ward, 1998; Margolis & Resnick, 2000): normalization versus equalization. The first hypothesis perceives that the adoption and diffusion of the internet among political actors do not seriously change the traditional hierarchy between major and minor players. The second approach emphasizes the internet's potential power that can level the playing field. In short, whereas the offline competitiveness of major actors over minor actors are simply replicated in cyberspace from the perspective of normalization, the proponents of equalization consider that "the internet clearly offers smaller political players a much cheaper means of reaching vast news audiences and, more importantly, contributes to the reformation of the political structure" (Margolis, Resnick, & Levy, 2003, p. 58).

As Hindman (2008) criticized in his book, *The Myth of Digital Democracy*, one of the problems is the limitation in the use of organizational websites (e.g., political parties, civic and advocacy groups, the mass media, and governments) in the analysis of either normalization or equalization effects of the internet. That is, party competition and campaigning online have been relatively well-documented in previous research (Ducke, 2004; Gibson, Nixon, & Ward, 2003; Hooghe & Teepe, 2007). On the other hand, the online visibility of individual politicians has been neglected. This provides the key motivation for this present study.

Another problem is that most of the previous studies have analyzed the visibility and competitiveness of party websites by examining the amount of user traffic and the existence, availability, functionality, and innovativeness of their content and service features and tried to determine whether the normalization or equalization mode is more feasible for a particular political system or whether both models can be applied. The present study proposes that the two models are very flexible and that this allows everything to be subsumed under them. This study argues that the political effect of the internet cannot be easily classified into one of the two models. Therefore, this study does not take any particular perspective.

2.2. Studies of web visibility

With the popularity of the internet, social and information science research has started to examine the concept of web visibility. The term "web visibility" generally refers to two things: a) the number of external web links an individual website has (InternetLab, 2005; Thomas & Willett, 2000), and b) the salience of particular people, objects, issues, and so forth on the internet (Ackland, Gibson, Lusoli, & Ward, 2010; Drèze & Zufryden, 2004; Khan & Park, in press; Kretschmer & Aguillo, 2004; Vaughan & Shaw, 2003, 2005).

Web visibility has been examined in several contexts. Martinez-Ruiz and Thelwall (2010) found that firms investing more in R&D and/or developing more high-impact technology are more likely to be visible on the internet in terms of the number of inlinks and web mentions, mainly because the market value of large firms tends to be higher than that of small firms. Park and Thelwall (2008) examined the online networking activities of politicians in terms of homepages and blogs and suggested that the position of politicians in a hyperlink network (i.e., visibility scores) tends to be closely related to their website content and/or services as well as to their attributes (e.g., their party affiliation, gender, and term). In a study of internet use by U.S. congressional members, Lipinski and Ndeddenriep (2004) found out that Republican members of the House of Representatives are more likely to be visible than other congressional groups in terms of the ownership of homepages.

Recently, a number of studies have examined online politics by measuring the web visibility of political actors. Gauvin (2010) examined the web visibility of candidates in the 2008 U.S. presidential election and suggested that web visibility indices can be used as a reliable indicator of political power and success. Lim and Park (2010) found a significant relationship between the web visibility of political candidates in the 2009 national by-election in Korea and their election results. Similarly, Hindman (2008) proposed that web visibility can be used as a method for identifying political power on the internet.

Further, Thelwall (2009) suggested that tracking web visibility by using search engines is a simple but powerful way to obtain valuable insights into the public's reaction to a topic or a person. Similarly, Gryc, Liu, Melville, Perlich, and Lawrence (2008, June) tracked blogs tagged with Barack Obama during the 2008 presidential election and examined the ways in which the internet played a political role in increasing his popularity. However, no study has systematically examined the relationship between the web occurrence of politicians and their attribute measures. In this regard, to develop a sophisticated theory about the political use of the internet, the present study

examines the ways in which certain politicians manage to appear frequently on the internet.

3. Research questions

This study examines the online performance of politicians by focusing on web mentions of their names and their offline attributes. Specifically, the study examines how the traditional socio-demographic background of politicians can impact the strength of their online presence. Thus, this study is guided by the following research questions (these questions are descriptive in nature, and given the discussion about the political role of the internet, the questions should be considered as exploratory):

- RQ1. Who are the most visible congressional members in the Korean websphere?
- RQ2. Do the socio-demographic attributes of congressional members affect their web visibility?

4. Data collection and analytical methods

4.1. Research subject

This study focused on Korea's 18th National Assembly members, who were elected in April 2008. Although the number of the congressional members is officially 299, there were 292 incumbents at the time of data collection. A pilot study of online data, a qualitative reliability test, was conducted using a combined search query (i.e., the “name” of the politician and “의원; National Assembly Member”) that is commonly used in relevant research (i.e., Lee, Kim, Ahn, & Jeong, 2010; Lim & Park, 2010). All of the returned pages were checked to determine whether the search results for individual politicians included any irrelevant information over the two-week pilot study period. The data for seven members contained irrelevant information (e.g., different people with the same name or other literal meanings associated with their names), and thus, for data reliability and consistency, the data for these seven members were excluded from this study, which left 285 members for the final sample.

For these members, their demographic (their gender and age) and political (their term, constituency, and party affiliation) attributes were collected from the National Assembly's official website (<http://www.assembly.go.kr>) and each member's website. Table 1 shows the socio-demographic attributes.

4.2. Web visibility

Web visibility is operationally defined as the number of web mentions of each congressional member. The web mentions of each

member's name were collected using a Naver search tool based on the API (Sams et al., in press). Naver (<http://www.naver.com>) is Korea's top search engine (Alexa, 2010); it accounted for a dominant 71.9% market share in Korea in 2009 (Kim, 2009, September 29). Naver opened its API service to the public in 2006, and the reliability of Naver's API has been confirmed. According to Bar-Ilan (2005), an accessible API allows the customization and development of useful tools and interfaces based on the publicly available features of the search engine. Mayr and Tosques (2005, July) compared the results obtained using two Google interfaces—the Google API and the standard web interface Google.com—and argued that there would always be differences (although minimal) between API results and normal search results.

The web mentions of each politician were collected across many platforms, including blogs, images, news, and websites. The numbers of hits retrieved using a combined search query (i.e., the “name” of the politician and “의원; National Assembly Member”) were collected on December 30, 2009.

5. Results

5.1. Descriptive information

Table 2 presents the online presence, including blogs, images, news and websites, of the 285 members. The means of online platforms were less than the standard deviations: blogs, 3122.47 ($SD = 4615.38$); images, 101.5 ($SD = 147.09$); news, 6269.79 ($SD = 9004.27$); and websites, 77,633.01 ($SD = 108,892.52$). The skewness values were positive, indicating a strongly right-skewed property. In addition, the kurtosis values were positive, indicating a sharper peak and a longer tail for their distributions than for a normal distribution. The results suggest that the web visibility indices for Korea's National Assembly members may follow a power-law distribution.

5.2. Correlations among web visibility indices

To examine the relationships among the four types of web visibility indices, the data were analyzed using Pearson and Spearman correlation coefficients. Although the Pearson correlation is the most efficient estimator for the normal distribution, the results may be distorted by outliers. For skewed distributions like those for the web visibility data, the Pearson correlation's lack of statistical robustness and efficiency can be problematic (Croux & Dehon, 2010; Morgenthaler, 2007). Recently, Croux and Dehon (2010) suggested that nonparametric correlation measures are statistically robust and efficient in terms of outliers. For this reason, we used the Spearman correlation, a nonparametric correlation measure, to confirm Pearson correlation values.

As shown in Table 3, both results were significant ($p < .01$) and positive (greater than .70). Specifically, in terms of the Pearson correlation coefficients, the highest was between blogs and news ($r = .963$, $p < .01$), followed by blogs and websites ($r = .934$, $p < .01$) and news and websites ($r = .912$, $p < .01$). In terms of the Spearman correlation coefficients, the highest was between blogs and news ($\rho = .955$, $p < .01$), followed by blogs and images ($\rho = .867$, $p < .01$),

Table 1
Socio-demographic background of South Korea's 285 National Assembly members.

Category		Count	%
Gender	Female	41	14.4%
	Male	244	85.6%
Age	Under 50	53	18.6%
	50–54	74	26.0%
	55–59	66	23.2%
	60–64	48	16.8%
	Over 64	44	15.4%
Term	First-term	125	43.9%
	Second-term	87	30.5%
	Three or more terms	73	25.6%
Constituency	Provincial region	129	45.3%
	National Capital region	105	36.8%
	Proportional representation	51	17.9%
Party affiliation	Opposition party	122	42.8%
	Ruling party	163	57.2%

Table 2
Descriptive statistics of web visibility indices.

	Blog	Image	News	Website
N	285	285	285	285
Mean	3122.47	101.50	6269.79	77,633.01
Median	2051	54	3734	48,065
SD	4615.38	147.09	9004.27	108,892.52
Skewness	7.30	3.71	5.48	6.11
Kurtosis	74.45	16.99	43.01	54.62
Minimum	380	7	395	3519
Maximum	57,915	1152	97,514	1,272,654

Table 3
Correlation among web visibility indices.

		1	2	3	4
1. Blog	<i>r</i>	–	.822**	.963**	.934**
	<i>rho</i>	–	.867**	.955**	.814**
2. Image	<i>r</i>		–	.838**	.783**
	<i>rho</i>		–	.864**	.734**
3. News	<i>r</i>			–	.912**
	<i>rho</i>			–	.773**
4. Website	<i>r</i>				–
	<i>rho</i>				–

Note. N = 285.

** $p < .01$ (two-tailed test).

images and news ($\rho = .864$, $p < .01$), and blogs and websites ($\rho = .814$, $p < .01$).

5.3. Top/bottom 10 politicians in terms of their overall web visibility

To rank the 285 congressional members in terms of the web visibility indices, we merged the online platform data sets by considering their contributions. We conducted a factor analysis to compute the weight (factor score coefficient) of each online platform data set. According to the Kaiser–Meyer–Olkin measure (.85) and Bartlett's test ($\chi^2 = 1665.84$, $df = 6$, $p < .01$), the factor analysis was appropriate for this purpose. Thus, only one factor was derived, which accounted for 90.73% of the variance. The factor loadings were .98 for blogs; .98 for news; .95 for websites; and .90 for images. In terms of the factor score coefficients, blogs were .27; images, .25; news, .27; and websites, .26. All the web visibility indices were weighted on the basis of these factor score coefficients. Then the four types of online platform data were combined as one variable. That is, the overall web visibility was a weighted sum accounting for the four online platforms. The average for the 285 congressional members was 22,969.17 ($SD = 32,065.53$).

Table 4 presents the top/bottom 10 congressional members in terms of the overall web visibility. In terms of the top 10, Geun-hye Park (376,804.10) showed the highest level of web visibility, followed by Sye-kyun Chung (166,961.02), Hoi Chang Lee (165,114.46), Mong Joon Chung (156,551.21), and Dong Young Chung (137,064.78). Two females and eight males represented the top 10. Their age ranged from 47 to 75. All of these 10 members served more than two terms. The constituency of four members was located in the national capital

Table 4
List of top 10/bottom 10 in terms of the overall web visibility.

	Visibility	Gender	Age	Term	Constituency	Party
Top 10						
Geun-hye Park	376,804.10	Female	58	4	Provincial	Ruling
Sye-kyun Chung	166,961.02	Male	60	4	Provincial	Opposition
Hoi Chang Lee	165,114.46	Male	75	3	Provincial	Opposition
Mong Joon Chung	156,551.21	Male	59	6	Capital	Ruling
Dong Young Chung	137,064.78	Male	57	3	Provincial	Opposition
Hyong O Kim	112,590.96	Male	63	5	Provincial	Opposition
Sang Soo Ahn	112,506.43	Male	64	4	Capital	Ruling
Joon Pyo Hong	107,661.65	Male	56	4	Capital	Ruling
Ki Kab Kang	103,486.98	Male	57	2	Provincial	Opposition
Kyung Won Na	88,185.61	Female	47	2	Capital	Ruling
Bottom 10						
Doo Ah Lee	1155.45	Female	39	1	Proportional	Ruling
Kum Lae Kim	2354.61	Female	58	1	Proportional	Ruling
Young Hee Jung	2733.34	Female	65	1	Proportional	Opposition
Ok Lee Kim	2814.14	Female	63	1	Proportional	Ruling
Jin Sam Lee	3254.55	Male	73	1	Provincial	Opposition
Choon Sik Lee	3396.38	Male	61	1	Proportional	Ruling
Yong Sam Lee	3497.13	Male	53	4	Provincial	Opposition
So Nam Kim	3773.46	Female	61	1	Proportional	Ruling
Gyu Back Ahn	3921.04	Male	49	1	Proportional	Opposition
Jae Jung Yoo	3932.65	Male	54	1	Provincial	Ruling

Table 5
Results of the median test and the Kruskal–Wallis test for gender differences.

	Gender	Frequency			Mean rank	Median test		Kruskal–Wallis	
		N	>M*	≤M*		χ^2	<i>p</i>	χ^2	<i>p</i>
Blog	Female	41	19	22	127.98	0.232	0.630	1.591	0.207
	Male	244	123	121	145.52				
Image	Female	41	20	21	134.50	0.009	0.924	0.509	0.475
	Male	244	121	123	144.43				
News	Female	41	19	22	125.27	0.232	0.630	2.217	0.137
	Male	244	123	121	145.98				
Website	Female	41	16	25	123.61	2.234	0.135	2.651	0.103
	Male	244	126	118	146.26				

Note. M* = Median.

region, and that of the remaining members, in provincial regions. There was no member who was elected through proportional representation. There were five ruling party members, leaving five for the opposition.

In terms of the bottom 10, Doo Ah Lee (1155.45) showed the lowest level of web visibility, followed by Kum Lae Kim (2354.61), Young Hee Jung (2733.34), and Ok Lee Kim (2814.14). The bottom 10 included five females and five males. Their ages ranged from 39 to 73. Except one member, all were serving their first term. Only three represented provincial regions, and the other members were elected through proportional representation. There were six ruling party members, leaving four for the opposition.

5.4. Demographic attributes

5.4.1. Gender

The Kruskal–Wallis test and the median test were performed to examine the gender difference in the web visibility index among various groups. As shown in Table 5, the difference was not significant for all online platforms ($p > .05$).

5.4.2. Age

The age of the 285 congressional members was categorized into five groups: under 50, 50–54, 55–59, 60–64, and over 64. As shown in Table 6, the results of the Kruskal–Wallis test and the median test indicate that there was no significant difference in the web visibility indices among the groups ($p > .05$).

Table 6
Results of the median test and the Kruskal–Wallis test for age differences.

	Age	Frequency			Mean rank	Median test		Kruskal–Wallis	
		N	>M*	≤M*		χ^2	<i>p</i>	χ^2	<i>p</i>
Blog	Under 50	53	21	32	136.04	2.822	0.588	0.805	0.938
	50–54	74	39	35	145.36				
	55–59	66	35	31	148.60				
	60–64	48	25	23	141.98				
	Over 64	44	22	22	140.13				
Image	Under 50	53	23	30	136.97	1.526	0.822	1.439	0.837
	50–54	74	39	35	147.84				
	55–59	66	35	31	149.70				
	60–64	48	23	25	140.28				
	Over 64	44	21	23	135.03				
News	Under 50	53	19	34	129.97	5.628	0.229	1.988	0.738
	50–54	74	37	37	141.58				
	55–59	66	36	30	149.53				
	60–64	48	27	21	145.98				
	Over 64	44	23	21	148.03				
Website	Under 50	53	24	29	131.81	1.897	0.755	3.207	0.524
	50–54	74	39	35	143.90				
	55–59	66	34	32	154.14				
	60–64	48	26	22	148.74				
	Over 64	44	19	25	132.00				

Note. M* = Median.

Table 7

Results of the median test and the Kruskal–Wallis test regarding the number of terms served.

	Term	Frequency			Mean rank	Median test		Kruskal–Wallis	
		N	>M*	≤M*		χ^2	p	χ^2	p
Blog	One	130	19	111	83.65				
	Two	89	66	23	186.73	118.773	0.000	132.691	0.000
	More**	73	61	12	209.37				
Image	One	130	25	105	92.53				
	Two	89	63	26	180.37	85.41	0.000	98.196	0.000
	More**	73	56	17	201.32				
News	One	130	13	117	76.62				
	Two	89	69	20	189.20	151.62	0.000	165.473	0.000
	More**	73	64	9	218.90				
Website	One	130	36	94	101.88				
	Two	89	59	30	175.60	46.847	0.000	66.666	0.000
	More**	73	51	22	190.48				

Note. M* = Median; More** = Three or more terms.

5.5. Political attributes

5.5.1. Consecutive term

The differences in the web visibility index among three groups (i.e., the first term, the second term, and the third term and above) were investigated using the Kruskal–Wallis test and the median test. There were significant differences among the groups in each of the online platforms (Table 7).

As a follow-up test, the Mann–Whitney U test was conducted to identify specific group differences. The web visibility indices for the first-term group were significantly different from those for the second-term group (blogs, $z = -9.47$, $p < .01$; images, $z = -7.84$, $p < .01$; news, $z = -10.47$, $p < .01$; and websites, $z = -6.67$, $p < .01$) and those for the group serving three or more terms (blogs, $z = -9.37$, $p < .01$; images, $z = -8.21$, $p < .01$; news, $z = -10.36$, $p < .01$; and websites, $z = -7.0$, $p < .01$). In addition, the web visibility levels for the second-term group were significantly different from those for the group serving three or more terms in terms of blogs ($z = -3.01$, $p < .01$), images ($z = -2.37$, $p = .02$), and news ($z = -3.96$, $p < .01$); the differences were nonsignificant in terms of websites ($z = -1.73$, $p = .08$).

5.5.2. Constituency

In terms of the political constituency, the 285 members were assigned into three groups: the national capital region, provincial region, and proportional representation groups. Table 8 presents the results of the Kruskal–Wallis test and the median test.

The web visibility indices were significantly different among the groups. In addition, the results of the Mann–Whitney U test indicate that the proportional representation group had significantly lower

Table 8

Results of the median test and the Kruskal–Wallis test regarding the constituency.

Constituency		Frequency			Mean rank	Median test		Kruskal–Wallis	
		N	>M*	≤M*		χ^2	p	χ^2	p
Blog	Provincial	129	74	55	152.18				
	Capital	105	56	49	159.82	17.556	0.000	31.094	0.000
	Proportional	51	12	39	85.16				
Image	Provincial	129	67	62	146.57				
	Capital	105	59	46	159.00	10.420	0.005	17.421	0.000
	Proportional	51	15	36	101.04				
News	Provincial	129	77	52	157.80				
	Capital	105	54	51	153.59	21.418	0.000	32.238	0.000
	Proportional	51	11	40	83.76				
Website	Provincial	129	67	62	156.24				
	Capital	105	61	44	151.25	13.315	0.001	23.506	0.000
	Proportional	51	14	37	92.53				

Note. M* = Median.

Table 9

Results of the median test and the Kruskal–Wallis test regarding the party affiliation.

	Party	Frequency			Mean rank	Median test		Kruskal–Wallis	
		N	>M*	≤M*		χ^2	p	χ^2	p
Blog	Opposition	122	73	49	157.70	8.552	0.003	6.787	0.009
	Ruling	163	69	94	132.00				
Image	Opposition	122	72	50	161.10	7.771	0.005	10.293	0.001
	Ruling	163	69	94	129.45				
News	Opposition	122	75	47	162.78	11.582	0.001	12.285	0.000
	Ruling	163	67	96	128.20				
Website	Opposition	122	68	54	161.18	2.983	0.084	10.380	0.001
	Ruling	163	74	89	129.39				

Note. M* = Median.

visibility levels than the provincial region group (blogs, $z = -4.90$, $p < .01$; images, $z = -3.40$, $p < .01$; news, $z = -5.44$, $p < .01$; and websites, $z = -4.60$, $p < .01$) and the national capital region group (blogs, $z = -5.31$, $p < .01$; images, $z = -4.04$, $p < .01$; news, $z = -4.94$, $p < .01$; and websites, $z = -4.25$, $p < .01$) for all online platforms. However, the differences between regions were insignificant (blogs, $z = -.70$, $p > .05$; images, $z = -1.18$, $p > .05$; news, $z = -.38$, $p > .05$; and websites, $z = -.50$, $p > .05$).

5.5.3. Party affiliation

The web visibility indices for the ruling party members and the opposition party members were compared by using the Kruskal–Wallis test and the median test.

As shown in Table 9, the results of both tests indicate significant differences between the two groups in terms of blogs, images, and news. In addition, whereas the results of the Kruskal–Wallis test indicate significant differences in website visibility levels, those of the median test did not indicate significant differences between the groups, χ^2 ($df = 1$, $N = 285$) = 2.23, $p = .08$.

6. Discussion

This study investigates the web visibility of Korea's National Assembly members by using an API-based Naver search tool and systematically examines the relationship between the level of their web visibility and their socio-demographic attributes.

6.1. Who are the most visible congressional members in the Korean webosphere?

To address the first research question, we identified the most visible congressional members in the Korean webosphere by using the overall web visibility index and considering four online platforms: blogs, images, news, and websites. The top 10 members were very popular politicians. Specifically, Geun-hye Park, who showed the highest level of web visibility, was the chairperson of the Grand National Party from 2004 to 2006 and served four consecutive terms through provincial elections. According to a recent public opinion poll conducted by Korean Herald Business, a Korean newspaper, Geun-hye Park was the most influential politician in Korea (Shin, 2009, October 5).

Sye-kyun Chung, who was ranked second in the top 10, was the president of the Democratic Party and was serving his fourth parliamentary term. Hoi Chang Lee, who ranked third, was a competitive presidential candidate, garnering 38.7%, 46.2%, and 15.1% of the votes in 1997, 2002, and 2007, respectively. He was the president of the Liberty Forward Party and was serving his third term. Mong Joon Chung, the chairperson of the Grand National Party, ranked fourth. He was also the vice president of the Fédération Internationale de Football Association (FIFA) and a large shareholder in the Hyundai Heavy Industry Group. Chung was serving his sixth term. Dong Young Chung ranked fifth. He was a presidential candidate (the Democratic Party) in 2007; he came in second with 26.2% of the

votes. He was serving his third term. Hyong O Kim ranked sixth. He was the speaker in the first half of the 18th National Assembly and was serving his fifth term. Sang Soo Ahn, who ranked seventh, was the floor leader of the Grand National Party and was serving his fourth term. Joon Pyo Hong, who ranked eighth, was the previous floor leader of the Grand National Party and was serving his fourth term. Ki Kab Kang, who ranked ninth, was the president of the Democratic Labor Party and was serving his second term. Finally, Kyung Won Na, who ranked tenth, was a former spokesperson of the Grand National Party and was serving her second term.

By contrast, the members belonging to the bottom 10 were relatively unpopular. Most of these members were newly elected. However, Yong Sam Lee, who represented a sparsely populated region, was serving his fourth term. In addition, only three of the bottom 10 members won the regional election, and the other seven had retained their parliamentary seats as a result of the proportional representation system.

6.2. Do socio-demographic attributes of congressional members influence their web visibility?

To address the second research question, we examined the relationship between the level of the 285 members' web visibility and their socio-demographic attributes. Their gender and age were not significantly related to web visibility, which is generally consistent with the findings of [Lipinski and Naddor \(2004\)](#), who determined no significant difference in the online activity of U.S. senators by age. However, political attributes (e.g., their term, constituency, and party affiliation) were significantly related to web visibility.

Specifically, with respect to their parliamentary term, those serving three or more terms ranked higher in terms of the level of web visibility than the first-term and second-term members. The first-term members had significantly lower web visibility than the other groups. In terms of the members' constituency, the proportional representation group members ranked lower in terms of web visibility than those directly elected through local elections, even though the difference between provincial regions and the national capital region was nonsignificant. Finally, in terms of their party affiliation, the opposition party members ranked higher than the ruling party members. These results suggest that politicians' political attributes may have considerable influence on their web visibility.

6.3. Normalization vs. equalization

This study began with a criticism of previous research on the political impact of the internet. Conventionally, previous studies have focused on political organizations, not on individual politicians. In addition, they have analyzed political systems by rigidly adhering to two theoretical perspectives: normalization and equalization. To address the limitations of previous research, we examined the differences in Korean politicians' web visibility at the individual level by considering their sociopolitical attributes. In addition, we framed the results neither by normalization nor by equalization. Thus, the findings require a broader, more flexible application of the theoretical arguments.

An important finding is that the most visible politicians in the Korean webosphere had highly qualified political careers and prominence. That is, the congressional members with higher levels of web visibility were more powerful politicians who enjoyed traditional political leverage in the real world. Their levels of web visibility reflected their offline power. This was supported by the results of the analysis examining the relationship between the members' web visibility and political career. The results indicate that newly elected politicians had lower levels of web visibility than re-elected ones. The directly elected congressional members, who competed with other politicians for the seat, had higher visibility than

the indirectly elected ones (i.e., those elected through proportional representation). These findings suggest that the level of web visibility may reflect the traditional power structure among individual politicians, not the new political hierarchy. Thus, the normalization perspective can now be considered.

In the present study, the level of Korean politicians' news coverage was closely related to the level of their web visibility on various online platforms such as blogs, images, and websites. As [Tresch \(2009\)](#) mentioned, if the news coverage mirrors the offline political power structure, other visibility indices can be considered as a reflection of political reality. This reasoning provides support for the normalization approach.

On the other hand, another finding of this study requires the equalization perspective. In terms of the members' party affiliation, the web visibility of the ruling party members (who held an absolute majority in the National Assembly) was significantly lower than that of the opposition party members. In terms of the web visibility of the top 10 members, five were from the opposition party. In particular, Ki Kab Kang had a high level of web visibility, even though he was a very progressive politician and a member of an opposition party. The results indicate the equalization effect; that is, the internet may play a major role in changing the traditional political hierarchy.

However, the differences in the level of web visibility between the ruling party members and the opposition party members might have been due to their political careers.

As shown in [Table 10](#), whereas 51.5% of the ruling party members were serving their first term, 33.6% of the opposition party members were newly elected. Half of the ruling party members were new members, and thus, they had lower levels of web visibility than the re-elected ones. This might have adversely affected their web visibility. This suggests that differences in politicians' party affiliation may be due to differences in their offline power structure and that the normalization approach may be better than the equalization approach in explaining party differences.

Overall, these findings suggest that those politicians who build political power and influence through their political career in the offline world are more likely to be visible in the webosphere. That is, it is likely that the traditional power structure transfers to the online world. Although the results indicate a limitation of the equalization effect (i.e., the internet reforms the offline hierarchical structure of individual politicians), they provide support for the normalization argument; that is, the internet replicates the real world.

6.4. Limitations and future research

This study has some limitations. Although the study demonstrates a significant relationship between the web visibility of National Assembly members and their political attributes, it ignores some important factors such as political activities, leaderships, and campaign contributions. Some of the members had a competitive advantage because they were party leaders or members of multiple legislative committees. In addition, the amount of campaign contributions received from the public can represent the offline power of politicians in terms of public support. These may be important variables in the examination of the relationship between online and offline politics. To better understand the impact of digital media deployment on society, future research should consider more

Table 10

Cross table between the party affiliation and the parliamentary term.

	One-term	Two-terms	Three or more	Total
Opposition parties	41 (33.6%)	47 (38.5%)	34 (27.9%)	122 (100%)
Ruling party	84 (51.5%)	40 (24.5%)	39 (23.9%)	163 (100%)

comprehensive variables that can indicate the inherent advantage of politicians in the offline world.

An important limitation of this study is its cross-sectional research design. Web visibility is not fixed; it changes over time. Although this study employs recent web data, it does not sufficiently address the dynamics of web visibility. The results indicate a positive relationship between web visibility and political attributes, but it is difficult to make predictions. For this reason, future research should provide a better understanding of web visibility by considering changes in this relationship over time.

Another limitation is its quantitative approach. The members' web visibility was measured by counting the online articles mentioning their names. However, such an approach cannot be used to determine whether an online article is favorable or not. The quality of web content is also important in the investigation a politician's online reputation and power. Thus, future research should develop a mixed approach for measuring web visibility.

7. Conclusion: contribution and implication

Using API-based web mention data obtained from a search engine, this study measured the online visibility of Korea's congressional members. The results were analyzed using a number of socio-demographic attributes, including the members' gender, age, party affiliation, constituency, and parliament term. As a result, this study suggests that politicians who serve longer are more likely to be visible on the internet and that the political power structure in the online world is similar to the traditional power structure in the offline world. These findings contribute to the literature by providing a better understanding of the political influence of the internet. The results provide support for the findings of Schulz (2004), who suggested that the internet is not so different from the traditional media logics. Just as the traditional media coverage reproduces the existing political power structure (Sellers & Schaffner, 2007; Tresch, 2009), the level of politicians' web visibility can reflect the traditional political hierarchy. This may indicate that the internet has already assumed a central role in the mediatization of politics.

The results have some practical implication for politicians. Those politicians in leadership positions or those serving multiple terms are more likely to be on the list of the top 10 politicians in terms of web visibility. By contrast, those serving fewer terms and those with less experience are more likely to be in the bottom 10. Thus, politicians wishing to be more visible on the internet should place greater emphasis on political achievements in the real world. In sum, by using a web-based indicator (i.e., web visibility), this study clearly demonstrates the "ubiquitous presence" of politicians in Korea's digitalized society.

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